

Compound Semiconductors

KKKC6353 Nanoelectronic Devices · Lecture 4, Part 3 of 3

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Silicon is best only at being silicon

Silicon wins on **oxide, crystal size and cost**. On almost everything else, a compound does better.

Faster electrons

GaAs, InGaAs and InP: phone radios, radar, the fastest transistors.

Light

Every LED and laser is made of a compound.

High voltage

SiC and GaN: electric cars and small chargers.

But not logic

Why they never replaced silicon in processors is part of the story.

1

The compound semiconductor family

Mix the elements, choose the gap.

Two elements that average to silicon

group II	group III	group IV	group V	group VI
	B	C	N	O
Mg	Al	Si	P	S
Zn	Ga	Ge	As	Se
Cd	In	Sn	Sb	Te
Hg				

III–V (red): GaAs, InP, GaN … IV (blue): Si, Ge, SiC II–VI (green): CdTe, ZnSe …

Ga has 3 outer electrons and As has 5: on average 4 per atom, just like silicon

Figure made for this course.

Pair a group III atom with a group V atom and the crystal has, on average, **four bonding electrons per atom**, like silicon. GaAs, InP and GaN are the best known.

Alloys: choose the gap you want

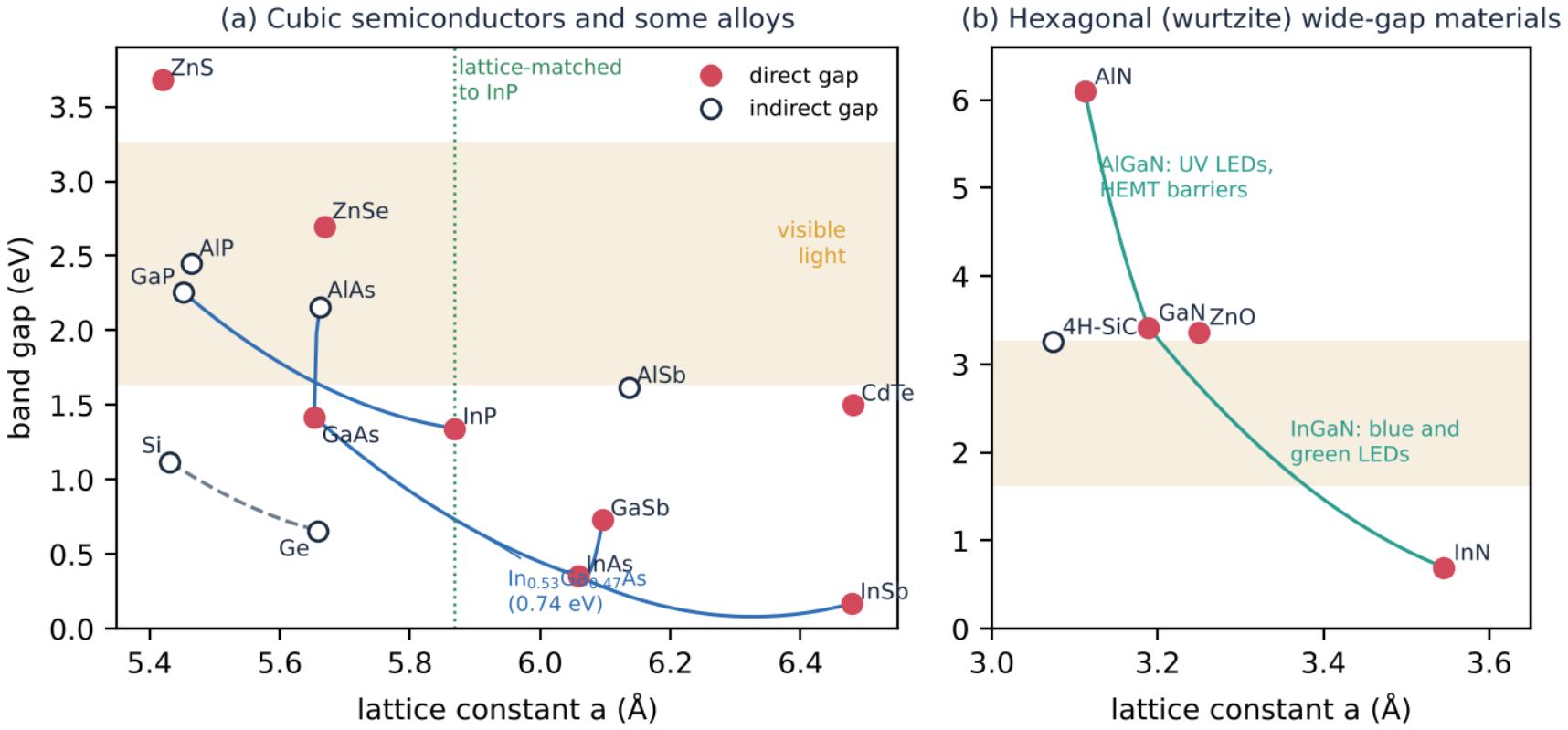


Figure made for this course; Ioffe database and Vurgaftman et al. (2001).

Mix two compounds and the gap changes **smoothly** with the mix. But layers grow cleanly on each other only if their **atom spacing matches**: AlGaAs on GaAs, InGaAs on InP.

2

Light and speed

Where the light and the fast electrons come from.

Why GaAs glows and silicon does not

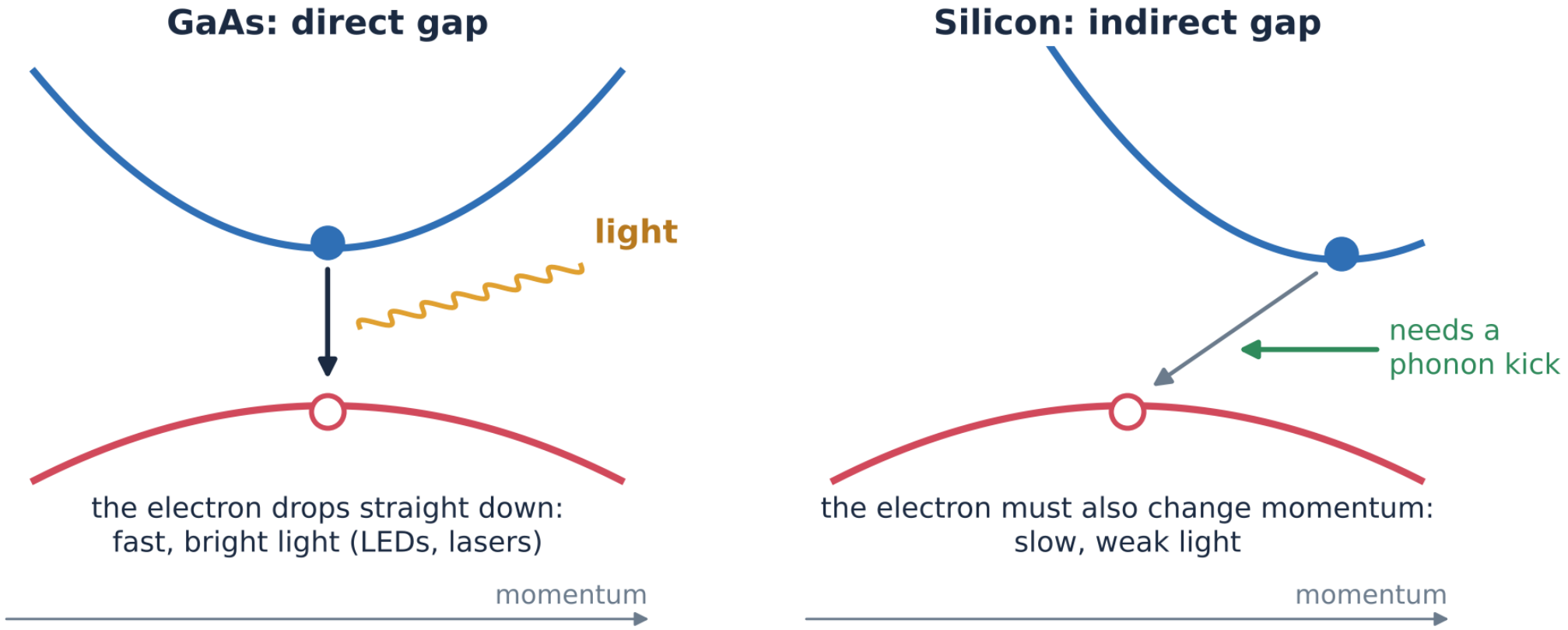


Figure made for this course.

A photon carries almost no momentum. In **GaAs** the electron drops straight down and emits light. In **silicon** it must also change momentum, which needs a phonon: light is weak.

Light electrons, and a speed limit

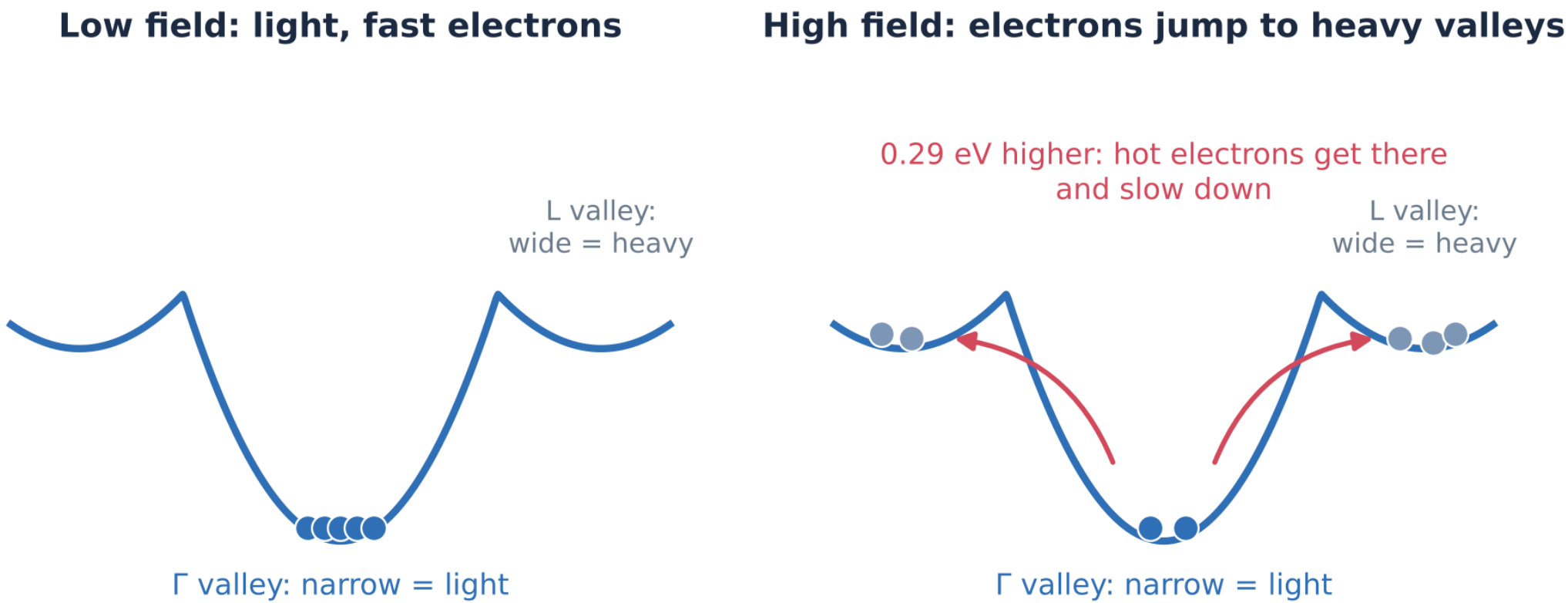


Figure made for this course.

GaAs electrons sit in a **narrow valley**: they are light and fast (mobility six times silicon's). At high fields they jump into **heavy valleys** and slow down.

The velocity peak

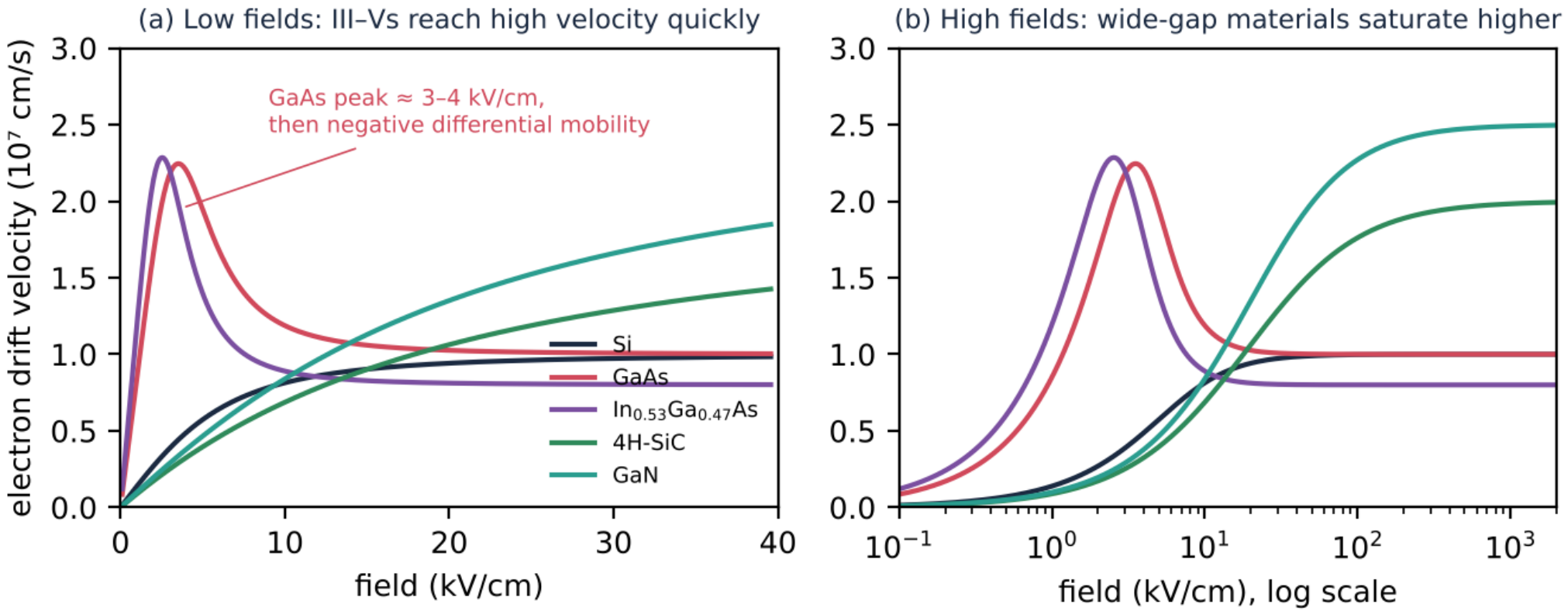


Figure made for this course from standard empirical fits.

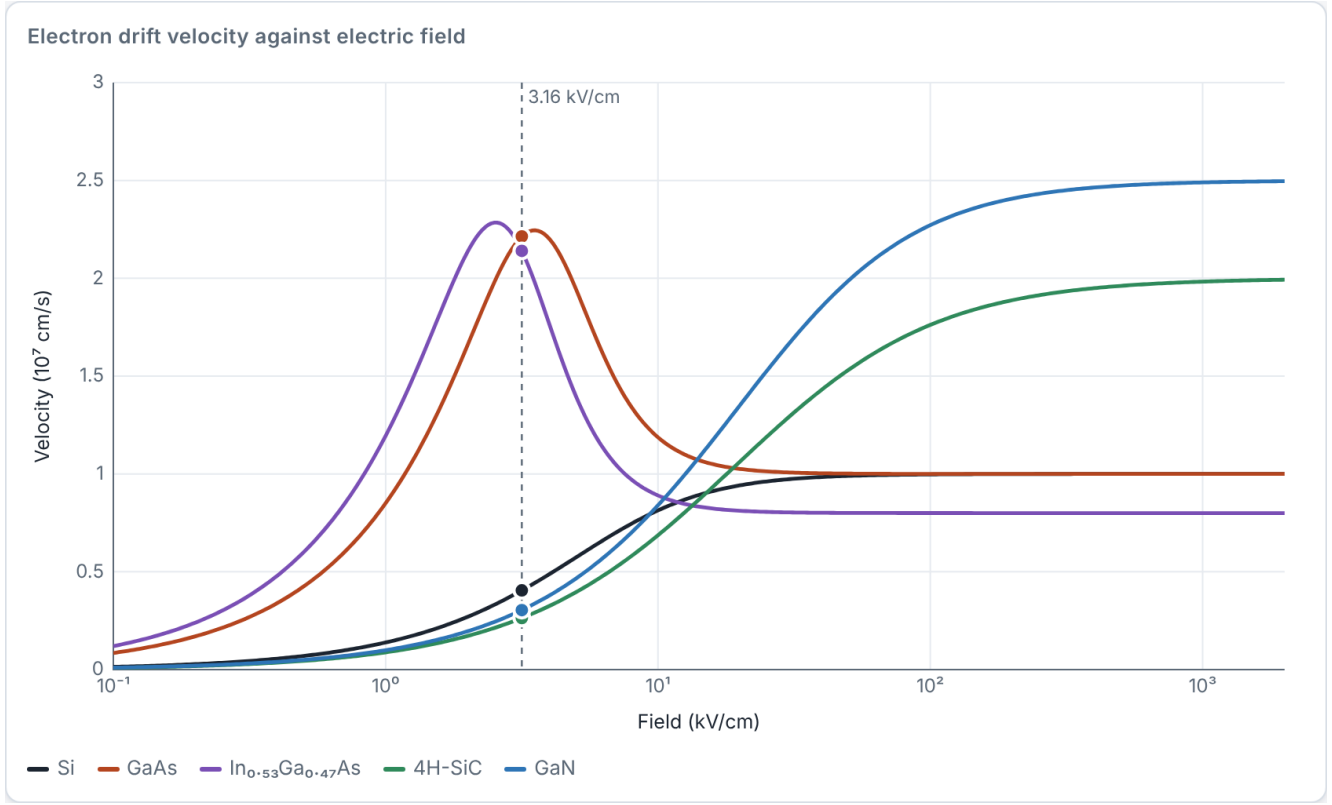
So the velocity **peaks** at a few kV/cm, about twice silicon's limit, and then falls back. The wide-gap SiC and GaN saturate later, but at a high speed.

Interactive animation 6

Compound semiconductors: velocity and band-gap map

faizverse.github.io/nanoelectronic-devices/L4/compound-semiconductors.html

- Velocity tab: compare Si, GaAs and InGaAs
- Switch to the log scale: where do SiC and GaN win?
- Map tab: walk along AlGaAs; where does the gap turn indirect?
- Find the InGaAs that matches InP, and its wavelength



3

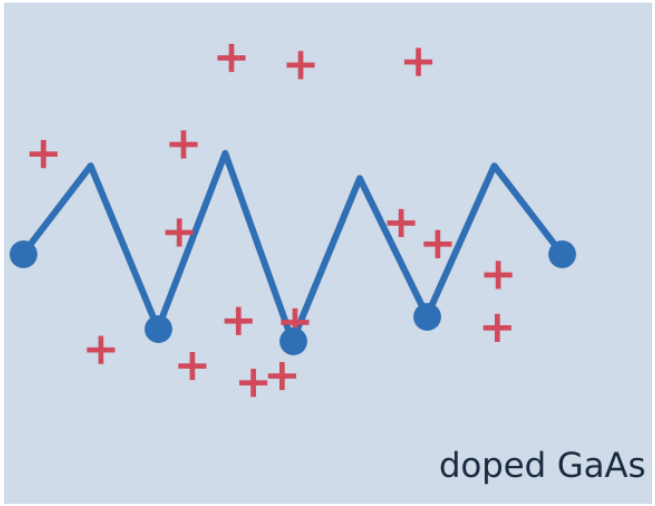
HEMTs and HBTs

Two transistors only heterostructures can make.

The HEMT: an express lane for electrons

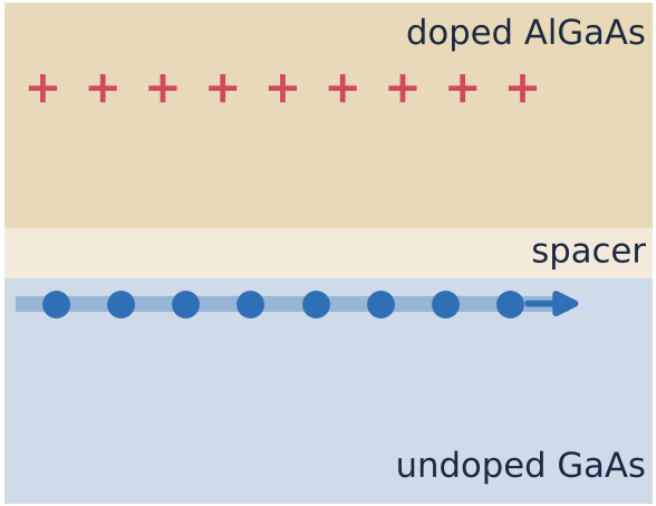
Doped channel: electrons share the lane with donors

collisions with ions: slow



HEMT: donors upstairs, electrons in a clean lane

2D electron gas: fast



GaN HEMT: no donors at all

polarisation charge at the interface pulls in 10× more electrons

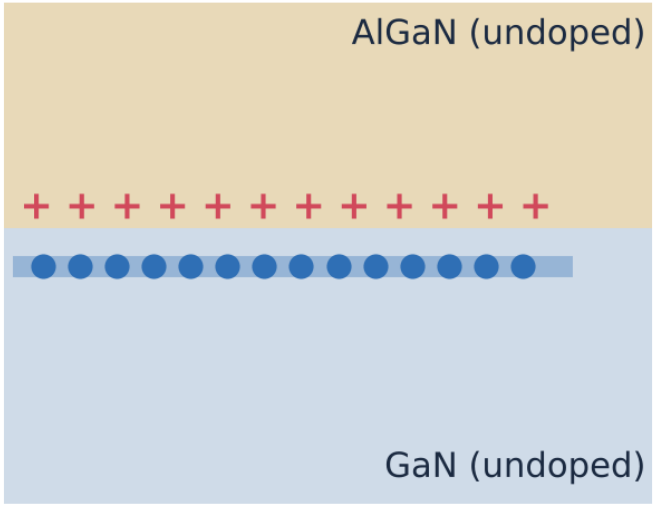


Figure made for this course.

Put the donors in a layer **above** the channel. Their electrons fall into clean GaAs and travel **without hitting the ions**. In GaN, polarisation alone supplies ten times more.

HEMTs in use

pHEMT

A strained InGaAs channel on GaAs: low-noise amplifiers and switches.

InP HEMT

The fastest transistors of all: amplification at **1 THz** (2015), used in radio astronomy.

GaN HEMT

5G base stations, radar, and fast power switches.

The HBT: a wall that stops the holes

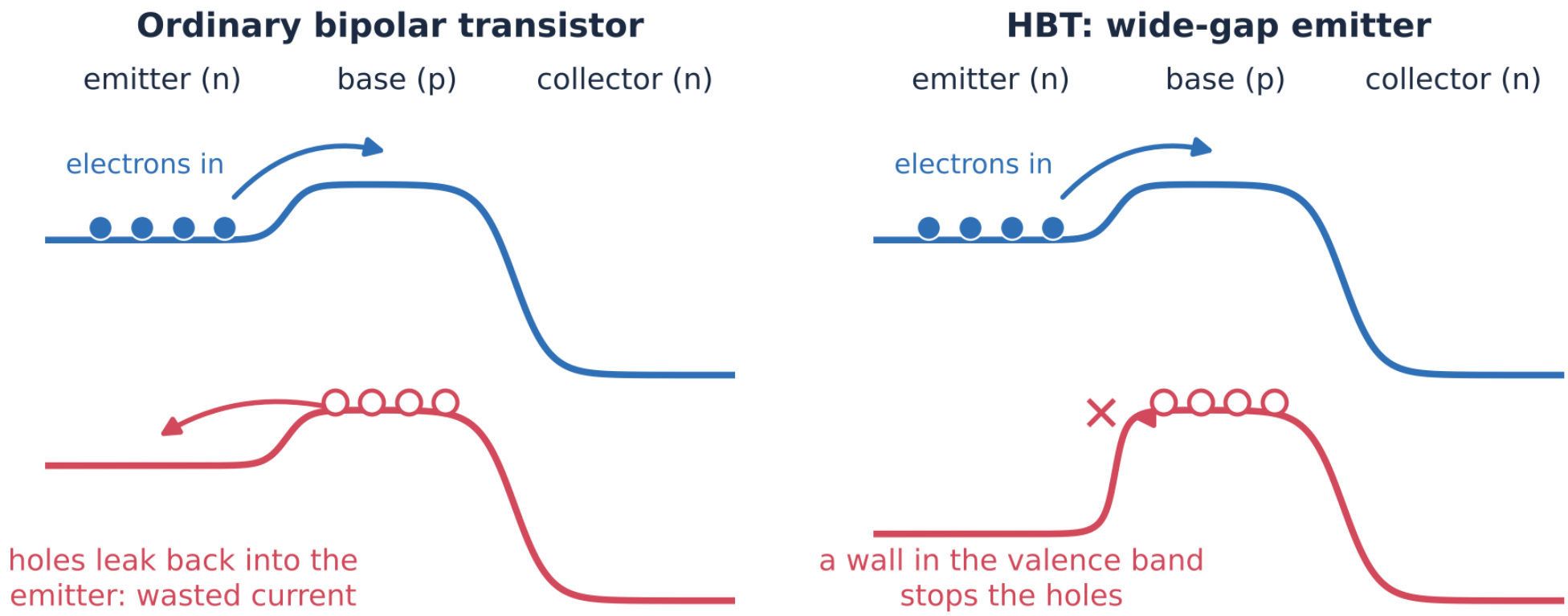


Figure made for this course.

In a bipolar transistor, holes leaking back into the emitter waste current. A **wider-gap emitter** puts a wall in their way, so the base can be heavily doped for **speed**.

4

Wide-gap semiconductors for power

Why your phone charger got small.

A thinner layer blocks the same voltage

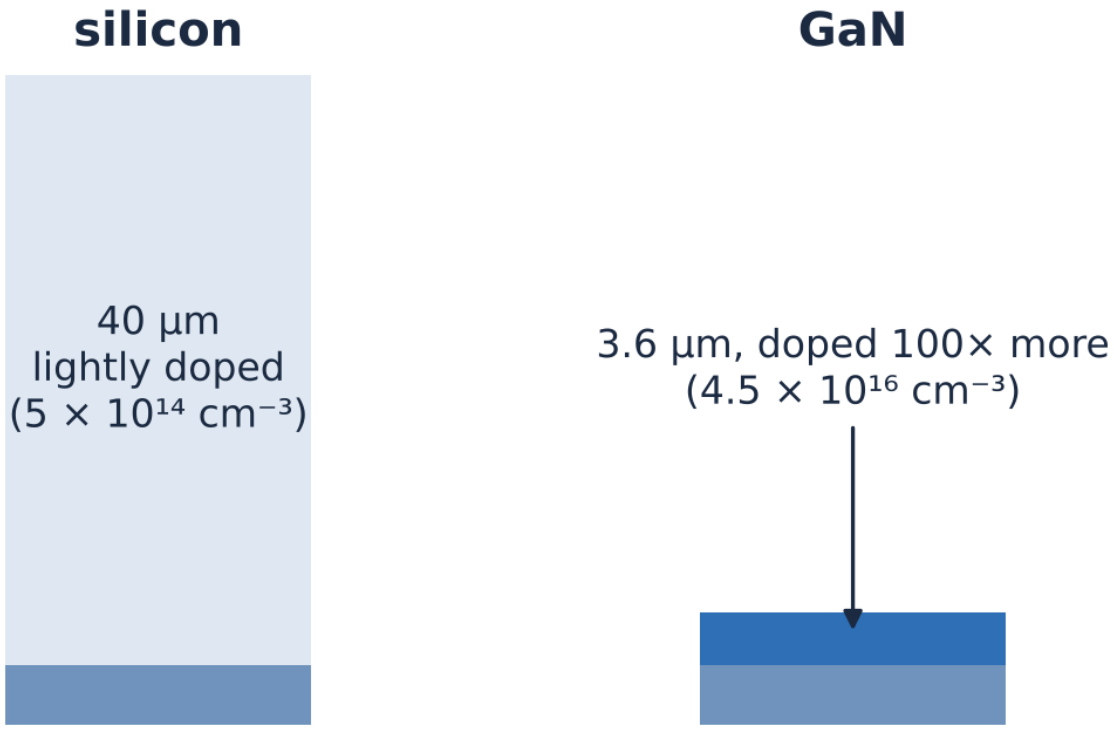


Figure made for this course; layers for 600 V drawn to scale from the ideal drift-layer design.

SiC and GaN withstand **ten times** silicon's field. The blocking layer can be ten times thinner and a hundred times more doped, so its resistance is **hundreds of times lower**.

SiC and GaN in 2026

SiC MOSFETs

650–1200 V: electric-car inverters, chargers, solar inverters.

GaN on silicon

100–650 V: USB-C chargers, data-centre power supplies.

Smaller everything

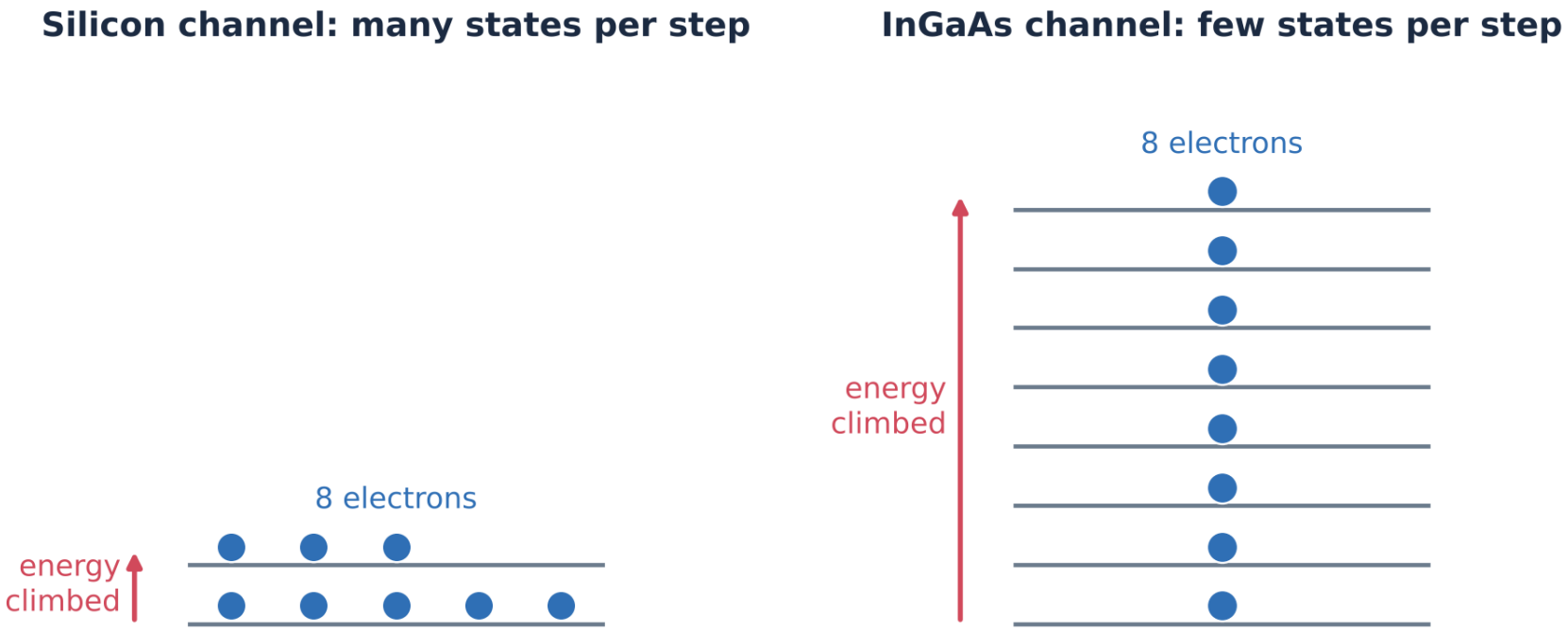
Smaller chips switch faster, so the inductors and capacitors around them shrink too.

5

Why not in processors?

Faster electrons, but fewer of them.

Light electrons, few states



The same 8 electrons must climb much higher in InGaAs. That climb is paid for with gate voltage, so each volt gives less charge.

Figure made for this course.

A light mass means **few states** at each energy. To hold the same charge, InGaAs electrons must climb higher, and that costs gate voltage. Faster electrons, but **fewer** of them.

The other obstacles, and where III–Vs went instead

Slow holes

CMOS needs fast p-channels too; III–V holes are no better than silicon's.

No good oxide

GaAs has no clean native oxide; deposited ones trap charge.

Beside silicon

III–Vs now join silicon for what it cannot do: GaN power and RF, lasers on silicon photonics.

6

Where compound semiconductors are used

Light, the highest frequencies, high power.

The places silicon cannot reach

- **GaAs**: phone power amplifiers, face-recognition lasers, red LEDs, space solar cells
- **InP**: lasers for optical fibre, the fastest radio chips
- **GaN**: blue and white LEDs, 5G and radar, chargers
- **SiC**: electric-car inverters, solar power

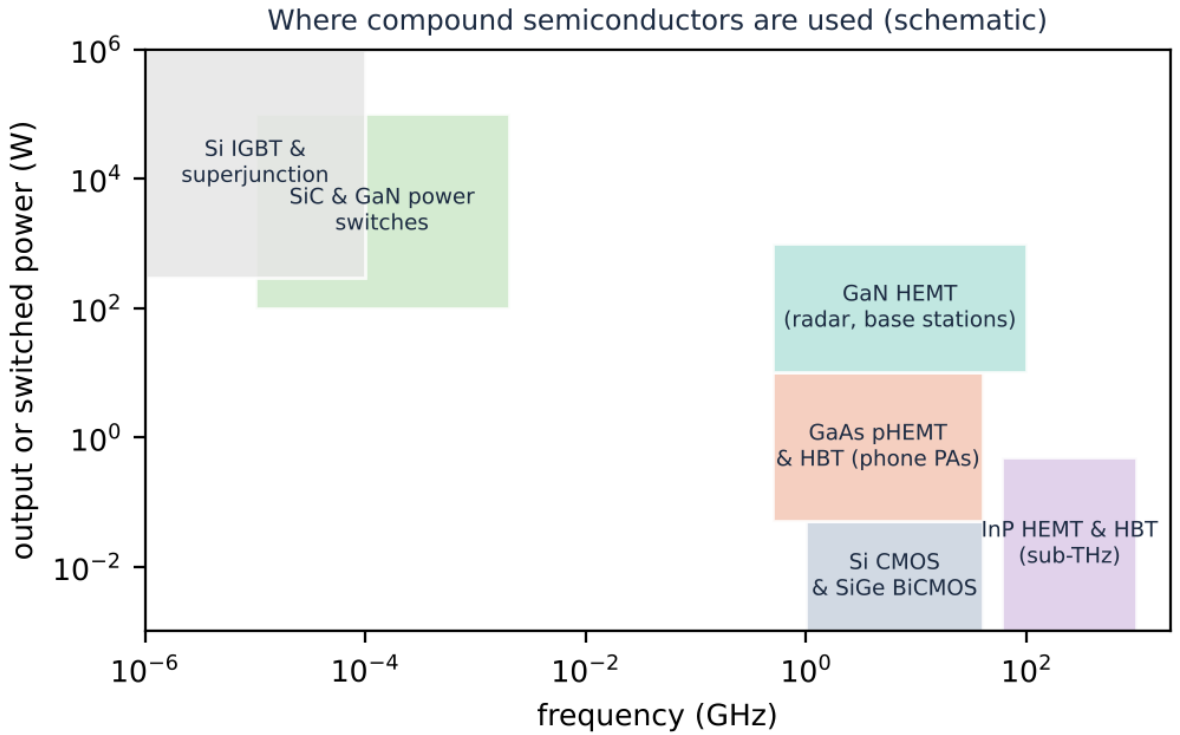
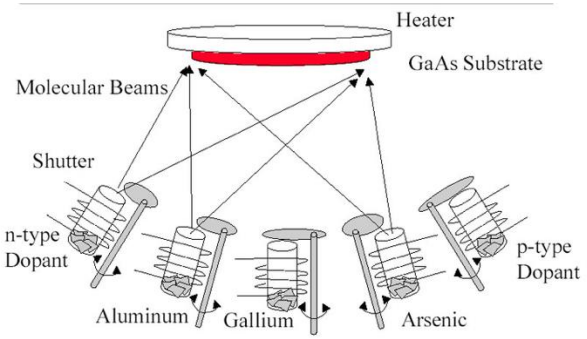


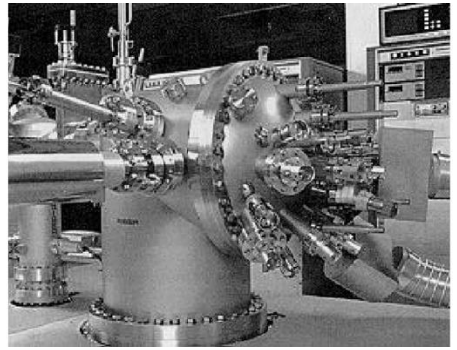
Figure made for this course; schematic boundaries.

How they are made

- Layers grown **one atomic layer at a time**: molecular-beam epitaxy (research) and MOCVD (production)
- Wafers are smaller and dearer than silicon: GaAs 150 mm, SiC 200 mm
- No native oxide, so gates are Schottky metals or heterojunctions



Reproduced from CSUN MSE-630 slides (old notes).



Supply and safety

Gallium is not rare

It is thousands of times more common than gold (correcting the CSUN deck). But it is only a **by-product** of aluminium and zinc refining, and China makes about **98%** of it, with export controls since 2023.

Arsenic is toxic

The arsine and phosphine gases used to grow III–Vs are extremely toxic: fabs use sealed gas cabinets, monitoring and abatement.

Part 3 in six points

1 Alloys

Mix III and V elements; choose the gap; match the spacing.

2 Light

Direct gaps emit light; silicon's indirect gap does not.

3 Speed

Light electrons are fast, until they jump to heavy valleys.

4 HEMT and HBT

An express lane for electrons; a wall for holes.

5 Power

Ten times the field: hundreds of times less resistance.

6 Not logic

Few states, slow holes, no oxide: used beside silicon.

Credits

Figures

Periodic-table, direct/indirect, valley, HEMT, HBT, drift-layer and density-of-states pictures, band-gap map, velocity–field and applications figures made for KKKC6353 (scripts in figures-source), using values from the Ioffe database and Vurgaftman et al. (2001). MBE schematic and photograph reproduced from CSUN MSE-630 slides in the old notes. Animation screenshot from faizverse.github.io/nanoelectronic-devices.

Sources

Sze and Ng, Physics of Semiconductor Devices (2007); Baliga, Fundamentals of Power Semiconductor Devices (2019); Kroemer (1957); Gunn (1963); Dingle et al. (1978); Mimura et al. (1980); Baliga (1982); Ambacher et al. (1999); Solomon and Laux (2001); Mei et al., IEEE EDL 36, 327 (2015); Infineon (2024–25); USGS Gallium (2025). Full list in L4 Document 4.